

PAWAN KUMAR RAI

Full-Stack Developer · ML Engineer · AI Automation · Data Analytics

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PROFESSIONAL SUMMARY

I see problems in three dimensions — rotating the challenge to find angles others miss. CSE undergraduate with 3+ internships, a published research paper at a national conference, and real products in production: a live revenue-generating web app with Razorpay integration, ML pipelines hitting 85%+ accuracy, and multi-LLM agentic workflows built from scratch. I don't just write code — I close the loop from data to model to user. Open to Full-Stack AI Engineer, ML Engineer, Data Analytics, or AI Automation roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

AI / ML: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Linear Regression, Random Forest, Gradient Boosting, Cross-Validation, R-squared, RMSE

AI Agents: n8n, LangChain, OpenRouter, Prompt Engineering, Multi-Agent Systems, Agentic Workflows, LLM CLI, Model Routing, ChatGPT, Claude, Gemini, Llama

Web Dev: REST APIs, Node.js, React, CSS Grid, Flexbox, Responsive Design, EmailJS, Razorpay, SEO Optimization

DevOps/Tools: Git, GitHub, Docker, Cloudflare Workers, GoDaddy DNS, Google Search Console, MySQL, Jupyter Notebook, Power BI

WORK EXPERIENCE

Data Science Intern — SkillCraft Technology (MSME | SCT/JUL25/4103) · Remote Jul 2025

- Accelerated insight turnaround across 4 real-world datasets by engineering end-to-end data pipelines — cleaning, EDA, and feature engineering in Python, Pandas, and Scikit-learn — cutting manual analysis effort by ~60% per dataset cycle.
- Produced visual analytical reports using Matplotlib and Seaborn to surface actionable, data-driven insights for stakeholders.
- Identified key behavioral patterns across titanic survival, iris classification, and traffic datasets — findings directly informed feature selection strategy for downstream model building.

AI / ML Intern — Edunet Foundation – AICTE · Remote Jun – Jul 2025

- Deployed production salary prediction model achieving 85%+ accuracy by benchmarking Linear Regression, Random Forest, and Gradient Boosting via k-fold cross-validation; evaluated with R-squared, RMSE, and MAE.
- Delivered full EDA with Seaborn visualizations enabling clear interpretation of salary drivers across experience, education, and role.
- Documented model performance comparison report with visual charts — enabling non-technical stakeholders to interpret salary drivers across experience, education, and role dimensions.

Web Developer Intern — Intern Pro · Remote Jun – Jul 2025

- Architected full authentication backend (MySQL schema, login/signup flow) for a team-built Portfolio Builder — enabling secure multi-user access and end-to-end deployment within one sprint.
- Improved cross-device accessibility by building fully responsive UI with CSS Grid and Flexbox across desktop, tablet, and mobile.
- Collaborated in a team sprint environment using Git version control — managing branch merges and resolving conflicts across a multi-developer codebase for production deployment.

RESEARCH & PUBLICATIONS

Facial Expression Recognition Using MediaPipe Landmarks and 3D-CNN Mar 2026

NCMPCS-2026, FOET, DSMNRU — National Conference on ML and Predictive Analytics Using Computational Sciences

- Proposed a real-time mental health monitoring system combining MediaPipe facial landmark extraction with a 3D-CNN architecture for expression classification; presented at NCMPCS-2026.
- Achieved robust multi-class expression recognition across 7 emotion categories by training on spatiotemporal facial landmark sequences using TensorFlow — addressing limitations of traditional 2D image-based approaches.
- Co-authored full research paper covering dataset preprocessing, model architecture design, training methodology, and evaluation metrics including accuracy and confusion matrix analysis.

PROJECTS

YouTube Data Analysis Dashboard · [Live: Railway](#) · Python, Pandas, YouTube API v3, Chart.js, Flask 2025

- Built end-to-end pipeline on YouTube India trending data — API ingestion, Pandas cleaning, NumPy analysis, 4+ Seaborn visualizations, Excel pivot reports via openpyxl.
- Deployed interactive Power BI dashboard (KPI cards, slicers) and web dashboard (HTML + Chart.js) with sentiment analysis using TextBlob.
- Identified trending content patterns across 500+ videos — revealing music and entertainment categories drove 3x higher engagement than news, producing actionable channel strategy insights.
- Built server.py (background Flask auto-trigger on page open) and update.py (one-command full pipeline) — deployed on Railway with GitHub auto-deploy; API keys secured via python-dotenv.

PawanPutra Travel · [Live: pawanputratravel.in](#) · TypeScript, JS, HTML5, CSS3, EmailJS, Razorpay 2025

- Built and deployed live car rental website with Razorpay payment gateway and EmailJS automation — received 10–15 real customer bookings.
- Boosted discoverability by architecting modular JS codebase (9 pages), configuring Cloudflare Workers edge caching and GoDaddy DNS — achieving full Google Search Console indexing.
- Implemented WhatsApp Business API integration and animated stats counter — improving user trust signals and increasing average session engagement on the landing page.
- Typed core booking and payment modules in TypeScript — improving code reliability and reducing runtime errors across the payment and form validation flows.

SalaryIQ · [Live: eemployee.streamlit.app](#) · Python, Scikit-learn, Pandas, Streamlit 2025

- Built full-stack ML web app on UCI Census Income data supporting 5 algorithms (Random Forest, Gradient Boosting, Logistic Regression, SVM, Decision Tree) with one-click training and configurable test splits.
- Engineered EDA dashboard with visualizations, batch CSV prediction with export, and single-entry prediction — end-to-end salary intelligence from upload to insight.
- Achieved 85%+ accuracy using Random Forest with feature importances; model config panel allows algorithm switching and hyperparameter tuning without code changes.

EDUCATION

B.Tech — Computer Science and Engineering | DSMNRU, Lucknow | SGPA: 7.8/10 Expected 2027

DSA, DBMS, Web Technology, Soft Computing, Digital Electronics, Discrete Mathematics

Class XII (CBSE) — DAV Public School, Kailash Hills, New Delhi | 75% 2022

Class X (CBSE) — DAV Public School, Kailash Hills, New Delhi | 89% 2020

CERTIFICATIONS & ADDITIONAL

Data Analysis with Python — IBM, 2024 | Python for Data Science & AI — IBM, 2024 | GenAI Powered Data Analytics — TATA/Forage, 2025 | HTML & CSS in Depth — Meta, 2024 | Programming with JavaScript — Meta, 2024 | Crash Course on Python — Google, 2023 | DSA in Python (20 hrs) — Udemy, 2024 | SQL for Data Analysis (7.5 hrs) — Udemy, 2024

Interests: Chess, Photography, Open Source, AI Automation